

Chapter 5

Price Controls and Market Efficiency

Chapter Outline/Learning Objectives

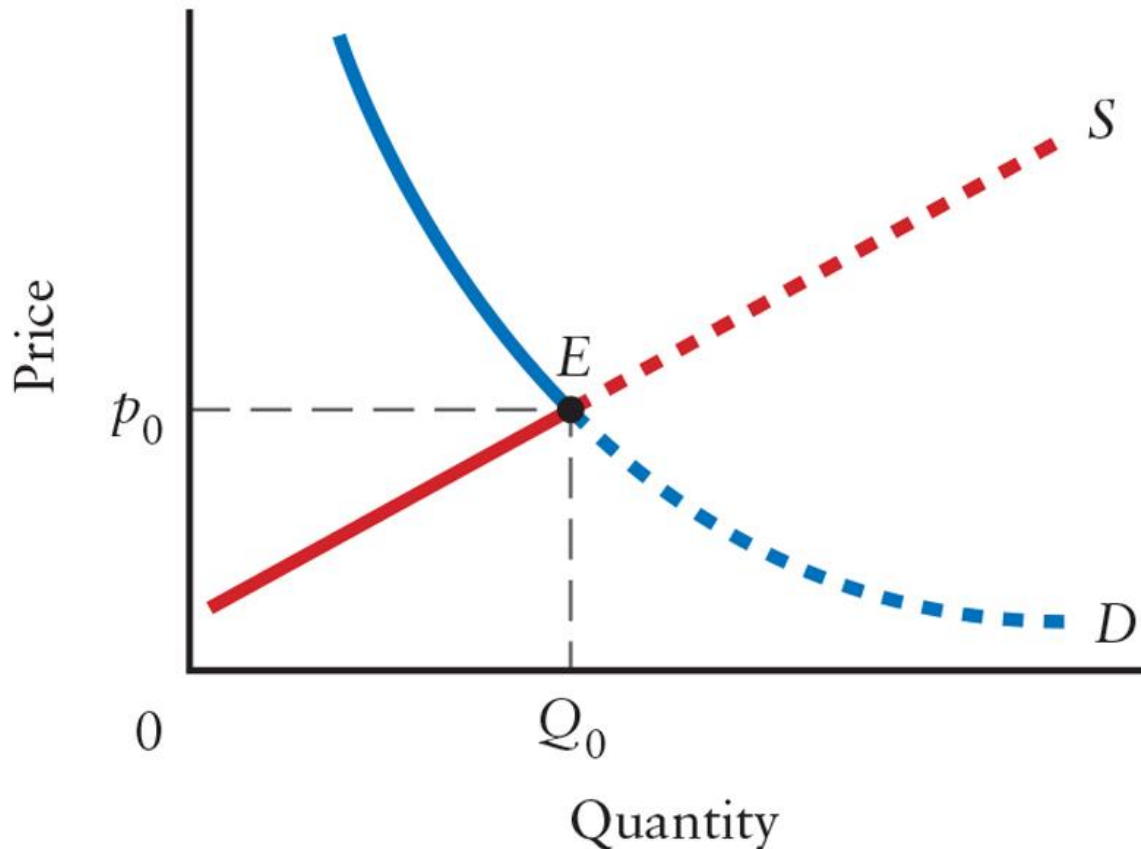
Section	Learning Objectives
	After studying this chapter, you will be able to:
5.1 Government-Controlled Prices	1. describe how legislated price ceilings and price floors affect equilibrium price and quantity.
5.2 Rent Controls: A Case Study of Price Ceilings	2. compare the short-run and long-run effects of legislated rent controls.
5.3 An Introduction to Market Efficiency	3. describe the relationship between economic surplus and the efficiency of a market. 4. explain why price controls and output quotas tend to be inefficient for society as a whole.

5.1 Government-Controlled Prices

- **Disequilibrium Prices**

- Voluntary market transactions require both a willing buyer and a willing seller.
- If the quantity demanded is less than quantity supplied, demand will determine the amount actually exchanged.
- If the quantity demanded exceeds quantity supplied, supply will determine the amount actually exchanged.
- In disequilibrium, the quantity exchanged is determined by the lesser of quantity demanded and quantity supplied.

Disequilibrium Prices



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Figure 5-1 The Determination of Quantity Exchanged in Disequilibrium

Price Floors

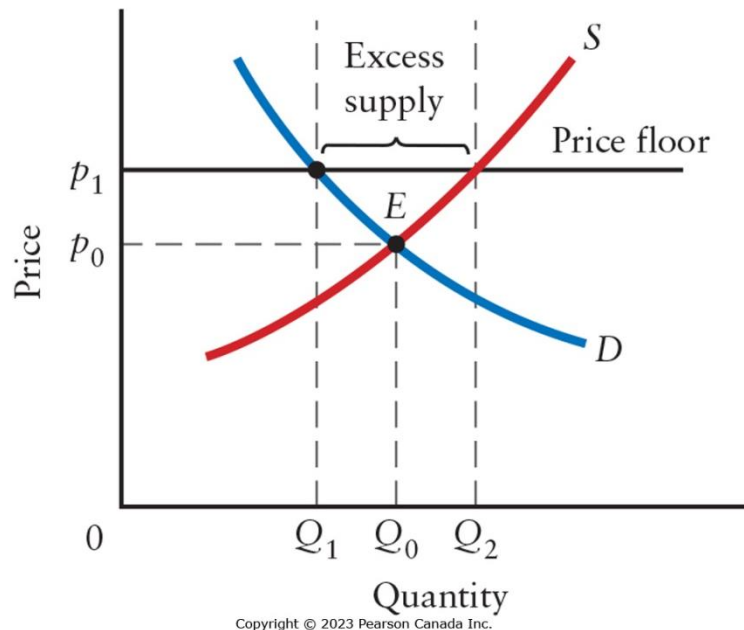


Figure 5-2 A Binding Price Floor

- A price floor is the **minimum permissible price** that can be charged for a particular good or service.
- A binding price floor leads to **excess supply**.

Apply Economic Concepts 5-1 (1 of 2)

- **Minimum Wages and Unemployment**

- A **minimum wage** is an example of a price floor in the labour market.
- In a *competitive labour market*, a binding minimum wage:
 1. Reduces the level of employment
 2. Increases quantity supplied of labour services
- **Unemployment increases.**
- The **owners of firms are made worse** off since they are now required to pay a higher wage than before the minimum wage was imposed.

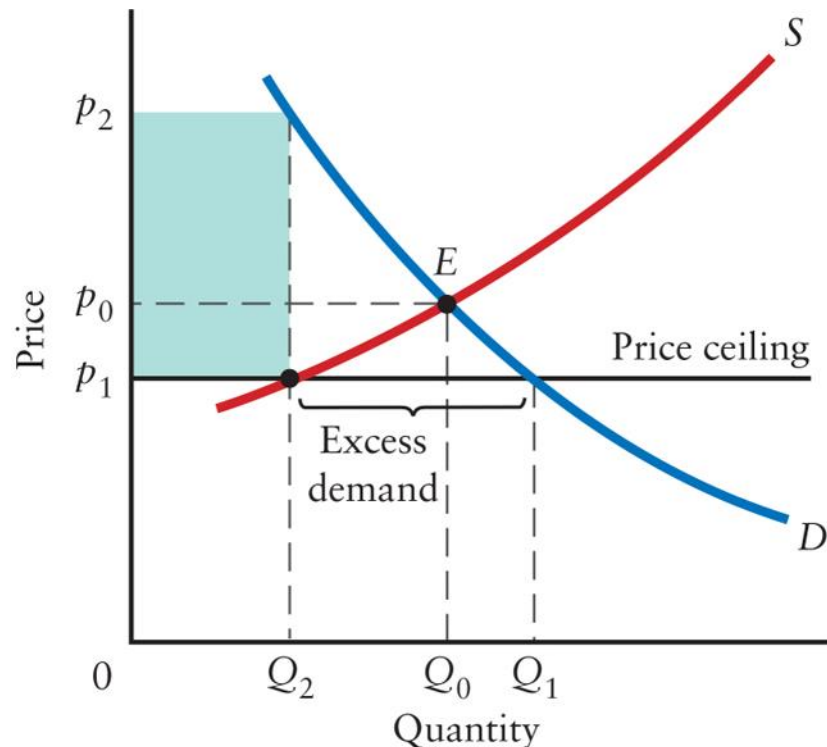
Apply Economic Concepts 5-1 (2 of 2)

- **Minimum Wages and Unemployment (continued)**
 - **Some workers gain** because they keep their jobs and they earn a higher wage rate.
 - Other workers lose because they lose their jobs as a result of the wage increase.
 - Minimum wage may have a different effect in *non-competitive* markets.

Price Ceilings (1 of 2)

- Free markets with flexible prices eliminate excess demand by allowing prices to rise.
- A price ceiling is a **maximum price**.
- With a binding price ceiling some other method of allocation must be adopted.
- First-come, first-served allocation results in buyers waiting in lines, disappointed to discover supplies are exhausted.
- May lead to a hidden market.

Hidden Markets



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- **Hidden market** – products are sold at prices that violate a legal price control.
- Profit can be made by buying at the controlled price and selling at the (illegal) hidden-market price.

Figure 5-3 A Price Ceiling and Black-Market Pricing

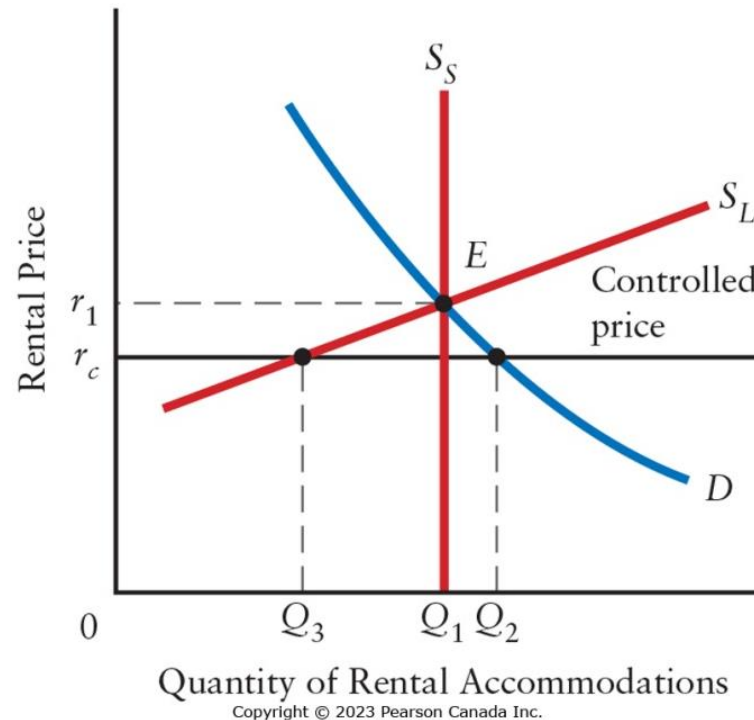
Price Ceilings (2 of 2)

- Goals of price ceilings are:
 1. To restrict production
 2. To keep specific prices down
 3. To satisfy notions of equity in the consumption of a product that is temporarily in short supply
- Hidden markets thwart government's objectives when imposing a price ceiling.

5.2 Rent Controls A Case Study of Price Ceilings

- **The Predicted Effects of Rent Controls**
 - Binding rent controls are a specific form of price ceiling, which have the following effects:
 - a shortage of rental housing because quantity demanded exceeds quantity supplied
 - alternative allocation schemes
 - hidden markets will appear

Figure 5-4 The Short-Run and Long-Run Effects of Rent Controls



Rent control causes housing shortages that worsen as time passes. The free-market equilibrium is at point E . The controlled rent of r_c forces rents below their free-market equilibrium value of r_1 . The short-run supply of housing is shown by the perfectly inelastic curve S_S . Thus, quantity supplied remains at Q_1 in the short run, and the housing shortage is Q_1Q_2 . Over time, the quantity supplied shrinks, as shown by the long-run supply curve S_L . In the long run, there are only Q_3 units of rental accommodations supplied, fewer than when controls were instituted. The long-run housing short-age of Q_3Q_2 is larger than the initial shortage of Q_1Q_2 .

Rent Controls in Ontario (1 of 2)

- Instituted in 1975
- Permitted increases in rents only where these were needed to pass on cost increases.
- 1990s – shortage developed in the rental-housing market (acute in Metro Toronto).
- Ontario government loosens rent controls by exempting some units and permitting increases when tenants vacated apartments.

Rent Controls in Ontario (2 of 2)

- 2017 – Ontario government expanded its rent controls generating considerable opposition.
- 2018 – Change of government led to exemption of newly constructed rental units from the rent control.

Who Gains and Who Loses?

- **Existing tenants** in rent-controlled accommodations are the principal gainers from a policy of rent control.
- **Landlords lose** because they do not receive the rate of return they expected on their investments.
- **Potential future tenants** also suffer because the rental housing they will require will not exist in the future.

Policy Alternatives

- Housing shortages can be reduced if the government, at taxpayers' expense, either **subsidizes** housing production or **produces** public housing directly.
- The government can make housing more affordable to lower-income households by providing income assistance directly to these households.
- Whatever policy is adopted has a resource cost.

5.3 An Introduction to Market Efficiency

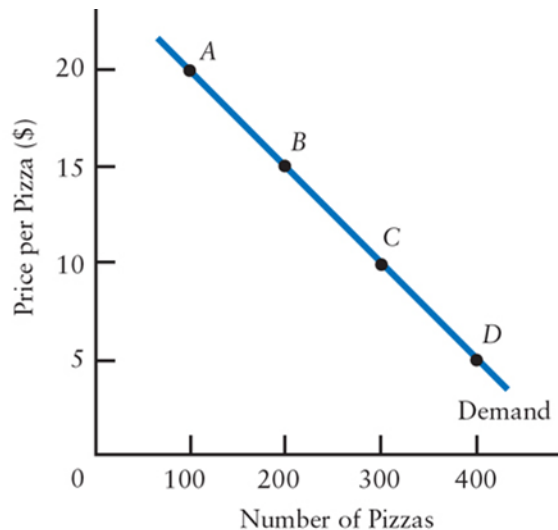
- The imposition of a controlled price generates benefits for some individuals and costs for others.
- Does a policy of legislated minimum wages make society as a whole better off because it helps workers more than it harms firms?
- Does a policy of rent controls make society as a whole better off because it helps tenants more than it harms landlords?
- Economists use the concept of ***market efficiency*** to address such questions.

Demand as “Value”

- The market demand curve for any product shows how much of that product consumers want to purchase.
- We can turn it around by starting with any given quantity and asking about the price.
- The demand curve tells us the highest price that consumers are willing to pay for a given unit.
- For each unit of a product, the price on the market demand curve shows the *value* to consumers from consuming that unit.

Reinterpreting the Demand Curve for Pizza

- For each pizza, the price on the demand curve shows the **value consumers receive** from consuming that pizza.



(i) Demand

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Figure 5-5 Reinterpreting the Demand and Supply Curves for Pizza

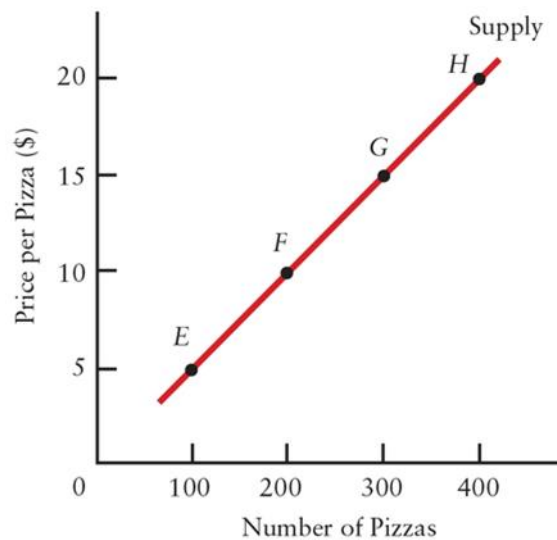
For each pizza, the price on the demand curve shows the **value consumers receive** from consuming that pizza; the price on the supply curve shows the **additional cost to firms of producing that pizza**. Each point on the demand curve shows the maximum price consumers are willing to pay to consume that unit. This maximum price reflects the value that consumers get from that unit of the product. Each point on the supply curve shows the minimum price firms are willing to accept for producing and selling that unit. This minimum price reflects the additional costs firms incur by producing that unit.

Supply as “Cost”

- The market supply curve for any product shows how much producers want to sell at each possible price.
- We can turn it around by starting with any given quantity and asking about the price.
- The supply curve tells us the lowest price that producers are willing to accept for a given unit.
- For each unit of a product, the price on the market curve supply shows the lowest acceptable price to firms for selling that unit. This lowest acceptable price reflects the additional cost to firms from producing that unit.

Reinterpreting the Supply Curve for Pizza

- For each pizza, the price on the supply curve shows the **additional cost to firms** of producing that pizza.



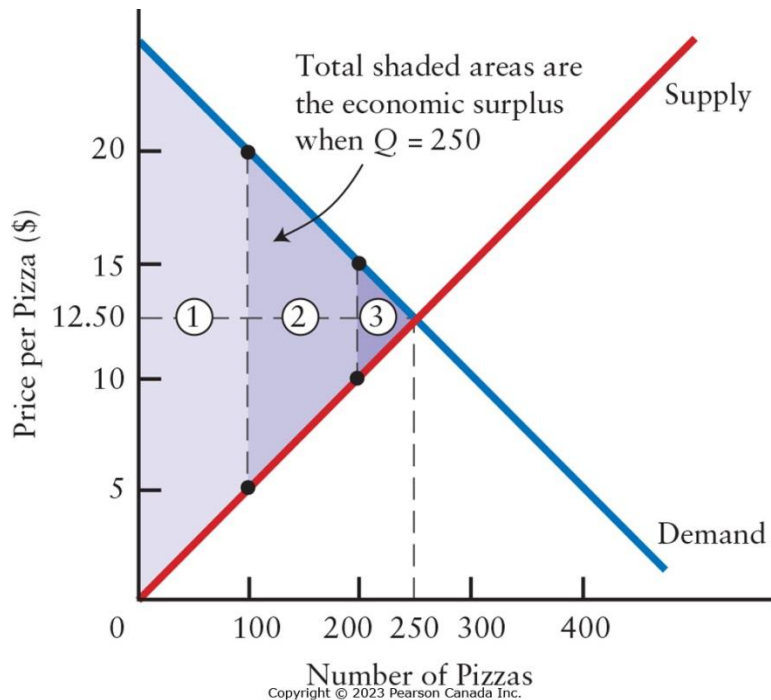
(ii) Supply

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For each pizza, the price on the demand curve shows the value consumers receive from consuming that pizza; the price on the supply curve shows the additional cost to firms of producing that pizza. Each point on the demand curve shows the maximum price consumers are willing to pay to consume that unit. This maximum price reflects the value that consumers get from that unit of the product. Each point on the supply curve shows the minimum price firms are willing to accept for producing and selling that unit. This minimum price reflects the additional costs firms incur by producing that unit.

Figure 5-5 Reinterpreting the Demand and Supply Curves for Pizza

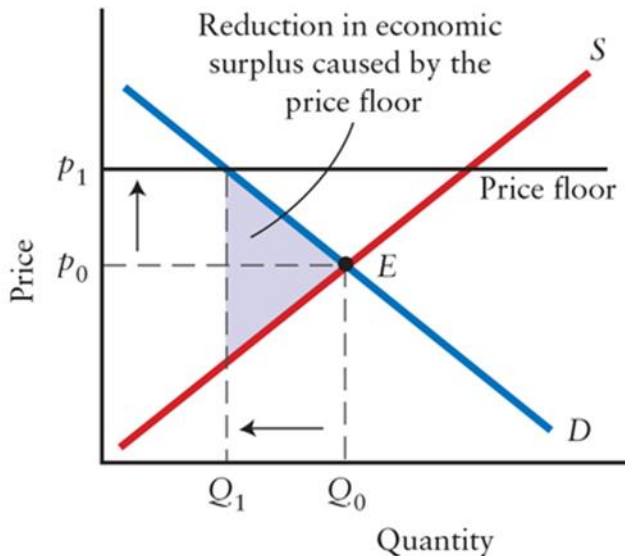
Economic Surplus and Market Efficiency



- Total economic surplus – **area below the demand curve and above the supply curve** – is maximized at the free-market equilibrium quantity.

Figure 5-6 Economic Surplus in the Pizza Market

Figure 5-7 Market Inefficiency with Price Controls (1 of 2)

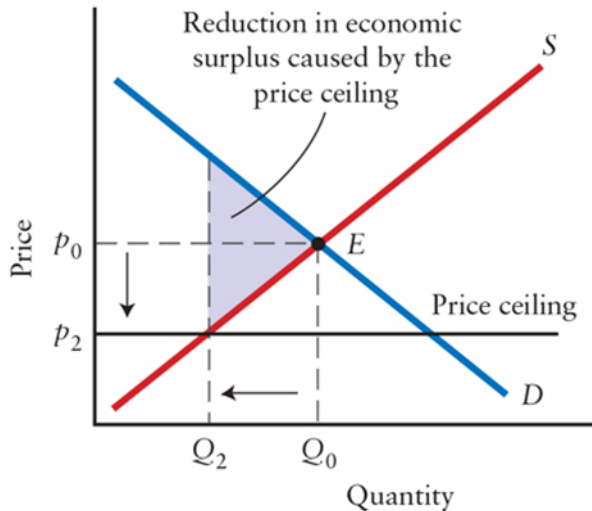


(i) Binding price floor

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- **Binding price floors in competitive markets lead to a reduction in overall economic surplus and thus to market inefficiency.**
- Production falls from Q_0 to Q_1 .
- With a free market, each unit of output from Q_0 to Q_1 generates economic surplus.
- The purple area shows **the deadweight loss**, which is the overall loss of economic surplus to society of the binding price floor.

Figure 5-7 Market Inefficiency with Price Controls (2 of 2)

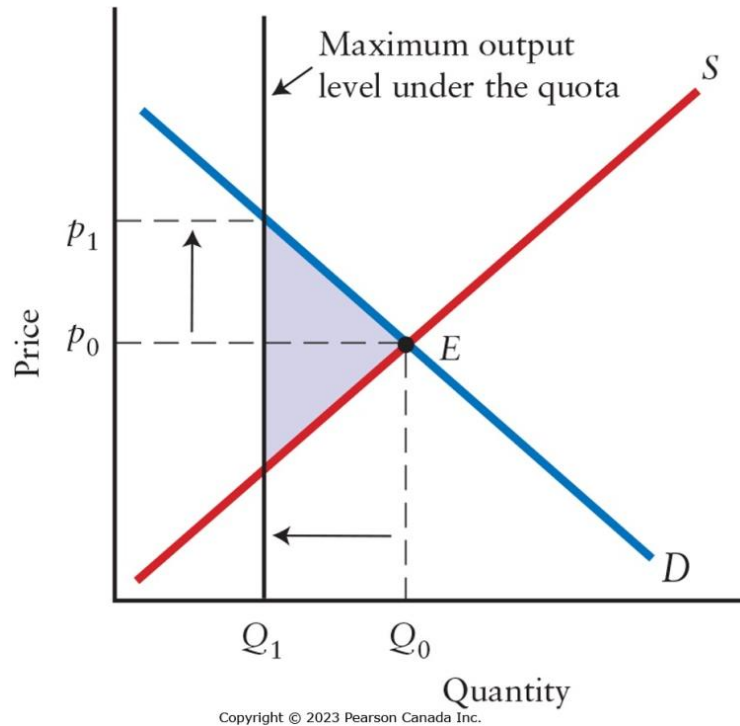


(ii) Binding price ceiling

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- **Binding price ceilings in competitive markets lead to a reduction in overall economic surplus and thus to market inefficiency.**
- Production falls from Q_0 to Q_2 .
- With a free market, each unit of output from Q_0 to Q_2 generates economic surplus.
- The purple area shows **the deadweight loss**, which is the overall loss of economic surplus to society of the binding price floor.

Figure 5-8 The Inefficiency of Output Quotas



- An output quota restricts output to Q_1 .
- The shaded area shows the reduction in overall economic surplus—the deadweight loss—created by the quota system.

Binding output quotas lead to a reduction in output and a reduction in overall economic surplus. The free-market equilibrium is at point E with price p_0 and quantity Q_0 . Suppose the government then restricts total quantity to Q_1 by issuing output quotas to firms. The market price rises to p_1 . The shaded area shows the reduction in overall economic surplus—the dead-weight loss—created by the quota system.

A Cautionary Word (1 of 2)

- Why does the government intervene in otherwise free markets when the outcome is inefficient?
- The answer in many situations is that the government policy is motivated by the desire to help a specific group of people.
- The overall costs are deemed to be a worthwhile price to pay to achieve the desired effect.

A Cautionary Word (2 of 2)

- Policymakers are making *normative* judgements.
- The job of the economist is to undertake *positive* analysis, emphasizing the *actual* effects of the policy rather than what might be *desirable*.
- The concept of economic surplus is not the only factor that the policy makers need to consider.
- Social responsibility is deemed to be more important to the conduct of society in markets prohibited by law.